

EE619: Assignment 2 – Inverter Simulations Submission Template

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Roll No. 230616

(Write your roll no. on space provided on all sheets. Penalty of -2 marks if the roll no. is missing.)

A1.

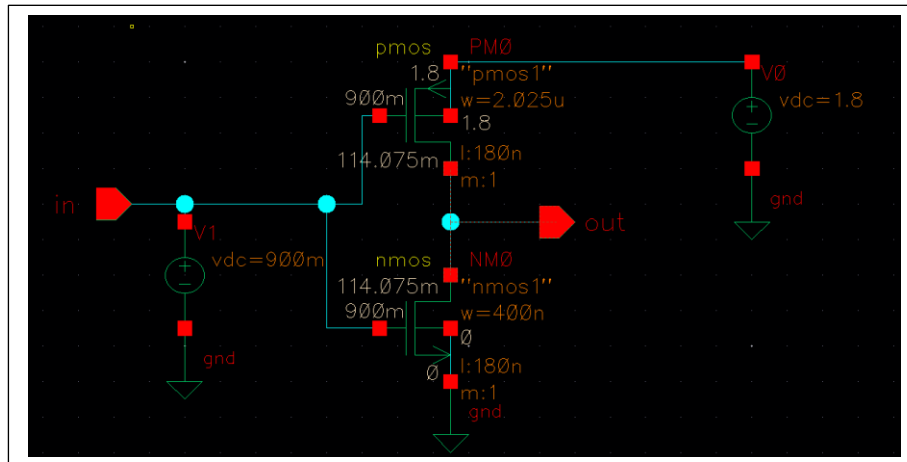


Fig. 1

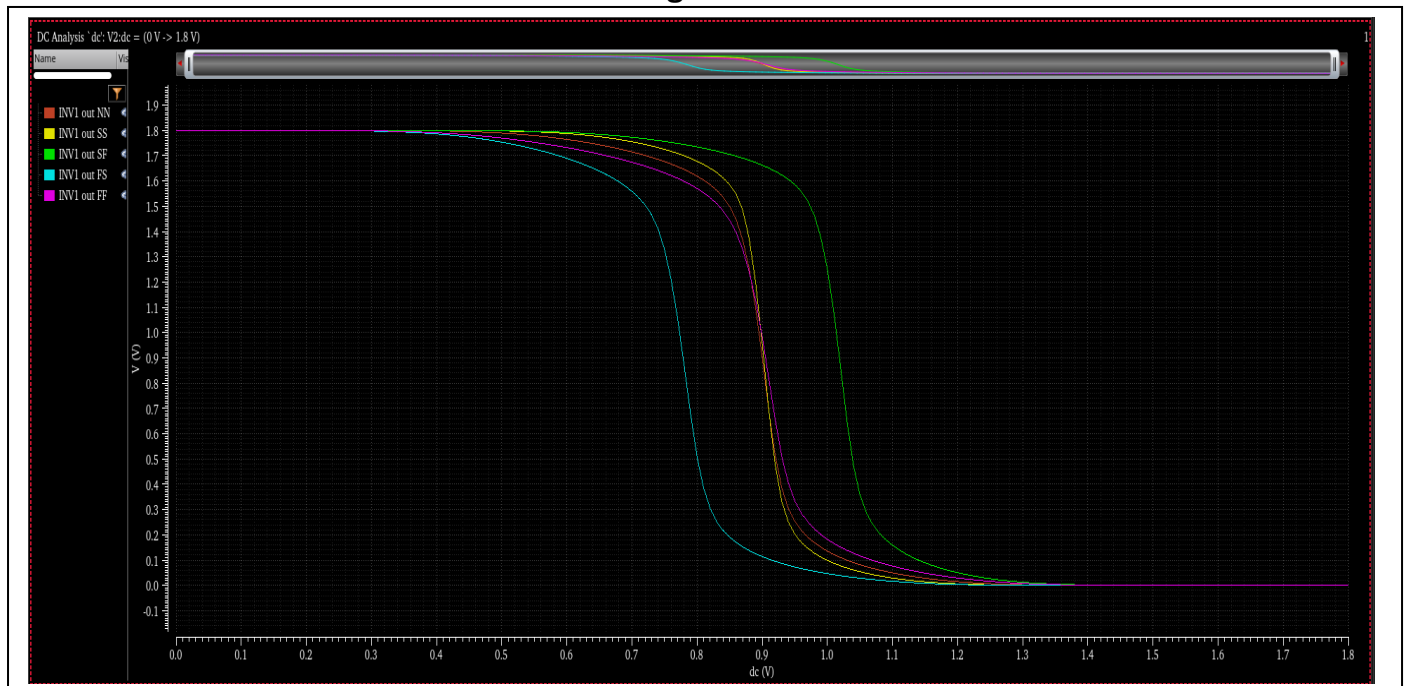


Fig. 2

Explanation-1:

	VIH	VIL	VOH	VOL
NN:	1.0345V	766.15mV	1.6606V	95.76mV
SS:	1.01925V	787.63mV	1.6918V	78.53mV
FF:	1.05485V	756.05mV	1.6257V	112.3mV
SF:	1.1554V	898.07mV	1.6644V	84.35mV
FS:	909.69mV	629.11mV	1.6622V	104.4mV

[INV1]

To find appropriate Wp I put fixed DC voltage 0.9 and varied Wp measured output and Choose wp where out is nearly 0.9. I Ran DC analysis where Input is varied from 0V to 1.8V and measured the output voltage for the varying input voltage. This is done for Inverter 1 with given specifications. Along with VTC I plotted derivative of output volt with respect to input volt and measured points at which it becomes -1 it occurred at two points of i/p voltage the lower one is VIL higher one is VIH. Then I used marker in VTC and checked at these two i/p voltage values and the two values of o/p voltages are obtained Lower one is VOL higher one is VOH.

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A2.

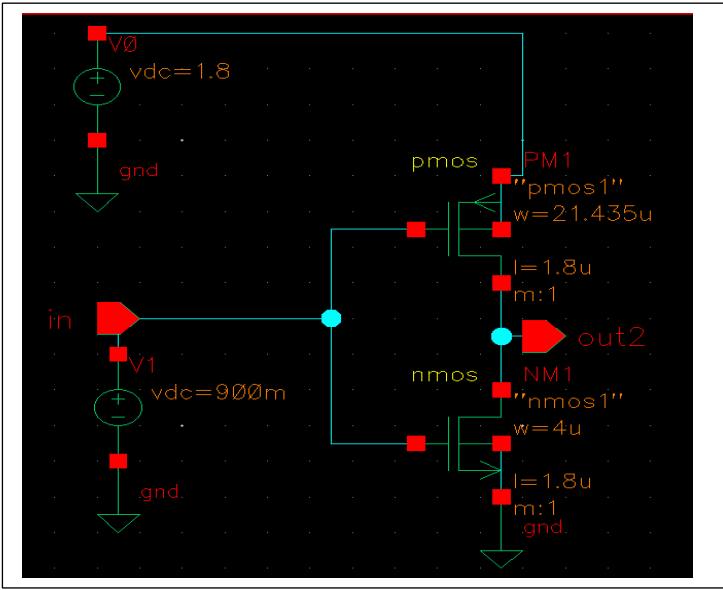


Fig. 3

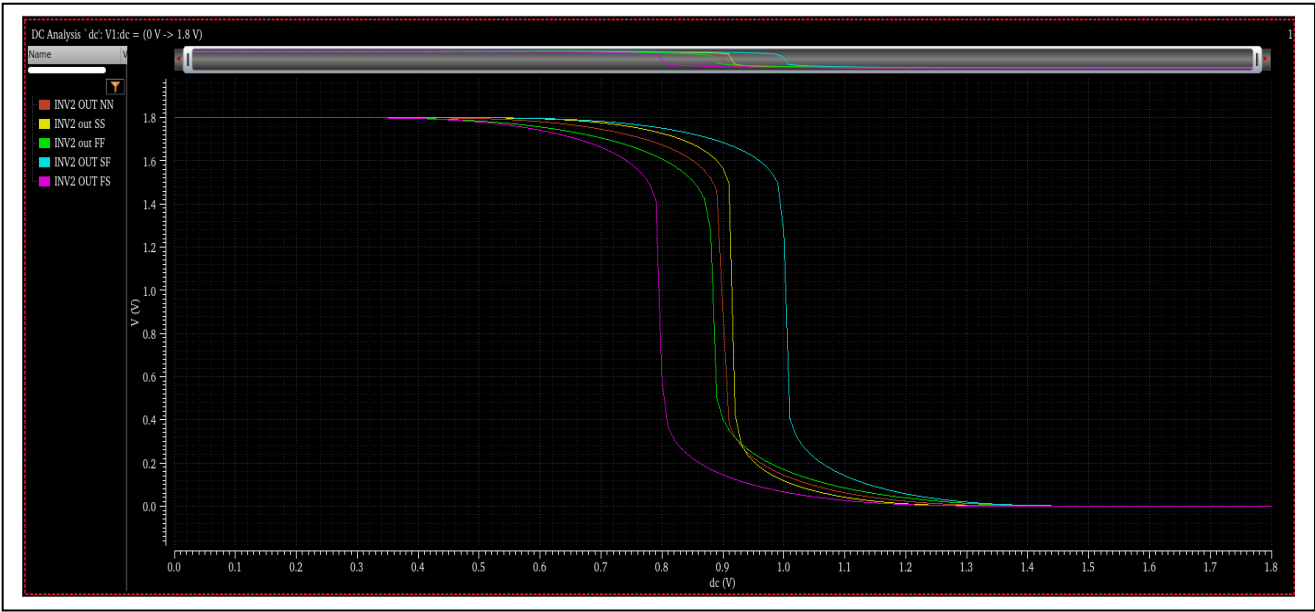


Fig. 4

Explanation-2:

VIH	VIL	VOH	VOL	
NN: 1.02516V	787.85mV	1.6856V	116.66mV	
SS : 1.02268V	832.3mV	1.6982V	93.385mV	
FF : 1.02705V	763.57mV	1.6527V	139.89mV	[INV2]
SF : 1.12879V	907.51mV	1.6769V	109.56mV	
FS : 920.95mV	688.02mV	1.6753V	112.75mV	

To find appropriate Wp I put fixed DC voltage 0.9 and varied Wp measured output and Choose wp where out is nearly 0.9. I Ran DC analysis where Input is varied from 0V to 1.8V and measured the output voltage for the varying input voltage. This is done for Inverter 2 with given specifications. Along with VTC I plotted derivative of output volt with respect to input volt and measured points at which it becomes -1 it occurred at two points of i/p voltage the lower one is VIL higher one is VIH. Then I used marker in VTC and checked at these two i/p voltage values and the two values of o/p voltages are obtained Lower one is VOL higher one is VOH.

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A3.

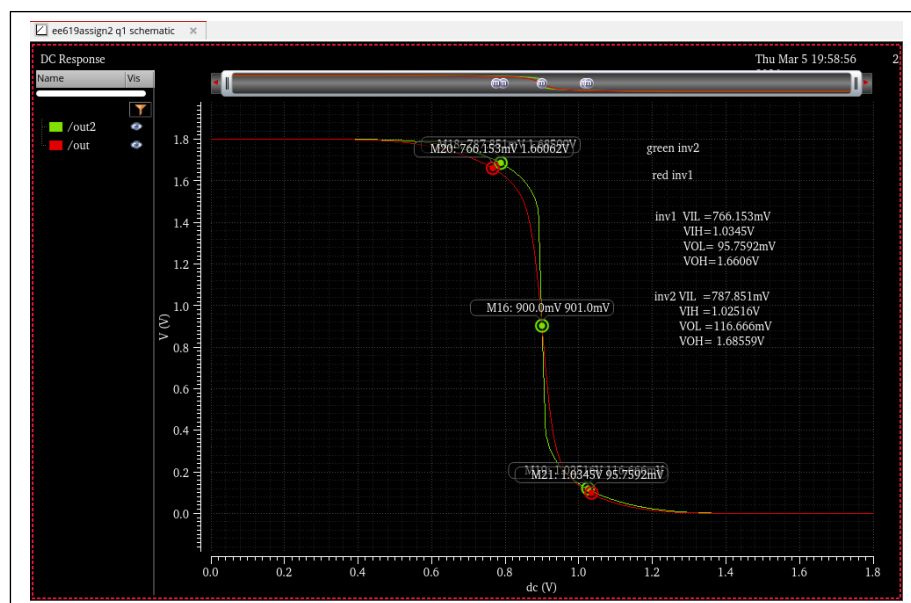


Fig. 5

Explanation-3:

The VTC curves of INV1 and INV2 were plotted on the same graph at the nn process corner and 27°C. inverter 1 has lower length (180nm) inverter 2 has higher length (1.8um) channel length modulation parameter (λ) is inversely proportional to length. Output resistance $[r_o = 1/(\lambda \cdot I_D)]$ is inversely proportional to λ . so output resistance increases with higher length (in both cases I_D is nearly same as W/L is kept nearly constant). High resistance implies high gain (g_m is nearly same in both cases) gain = $g_m (r_o)$. So that's the reason inverter 2 have steep slope compared to inverter1. V_m is designed to be at the $V_{DD}/2$ point hence both are at $V_{DD}/2$ approximately.

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A4.

Metric	INV-1	INV-2
V_{IH}	1.0345V	1.02516V
V_{IL}	766.15mV	787.85mV
V_{OH}	1.6606V	1.6856V
V_{OL}	95.76mV	116.66mV
Noise Margin (High)	0.6261V	0.66044V
Noise Margin (Low)	0.6704V	0.6712V
Noise Immunity	0.6261V	0.66044V

Table-1

Explanation-4:

The threshold values shown in Table-1 were obtained from VTC curves of both inverters at 27 °C and NN corner .

Using these threshold values, the noise margins of the inverter were calculated.

Noise Margin (High): $NMH = V_{OH} - V_{IH}$

This value indicates how much noise can be added to a logic '1' signal before the voltage drops below the minimum level that is still recognized as a valid logic high.

Noise Margin (Low): $NML = V_{IL} - V_{OL}$

This value represents how much noise can be added to a logic '0' signal before the voltage rises above the maximum level that is still recognized as a valid logic low.

To determine the overall noise immunity of the inverter, We use the smaller of the two noise margins . This value represents the maximum noise that the inverter input can tolerate without causing an incorrect logic output.

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A5.

Metric		TPHL	TPLH
Rise/Fall Time (ps)	10	356.78ps	215.7ps
	50	362ps	223.3ps
	100	368.5ps	233.2ps
	500	430.3ps	317.2ps
Load Capacitance (pF)	0.005	196.4ps	122.2ps
	0.1	362ps	223.3ps
	0.5	1.6854ns	1.0337ns

Table-2

Explanation-5:

From the simulation results in Table-2, the propagation delays of INV1 are much larger than the required delay of 2 ps under the given conditions. For example, when the load capacitance is 0.1 pF and the input rise/fall time is 10 ps, the delays are about 356.78 ps for TPHL and 215.7 ps for TPLH. It is also observed that when the input rise/fall time or the load capacitance increases, the delay increases. This is expected because the inverter has to charge and discharge the load capacitance through the PMOS and NMOS transistors, and this process takes a finite amount of time.

To check whether the delay could be reduced, the transistor widths were increased. When both PMOS and NMOS widths were increased to around 5 mm, the delays reduced to about 8.6 ps for TPHL and 20 ps for TPLH. Increasing the transistor width reduces the effective resistance of the devices, which helps the load capacitance charge and discharge faster.

However, even with such very large transistor sizes, the delay is still greater than the required 2 ps. Also, transistor widths of this order are not practical in real designs and would introduce large parasitic capacitances. Because of this, it is not possible to achieve a delay below 2 ps for this inverter just by sizing the transistors. The delay is mainly limited by the capacitance at the output node and the RC charging and discharging behavior of the circuit. It is possible when load cap is zero only parasitic cap is present.

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A6.

Metric	INV-1	INV-2
TPHL	362ps	634ps
TPLH	223.3ps	385.4ps

Table-3

Explanation-6:

The propagation delays of INV1 and INV2 were measured at the NN process corner and 27 °C with a load capacitance of 0.1 pF and an input rise/fall time of 50 ps. From Table-3, the delay for INV1 is about 362 ps for TPHL and 223.3 ps for TPLH. For INV2, the delays are higher, around 634 ps for TPHL and 385.4 ps for TPLH. This difference can be clearly observed from the values in the table. Although the (W/L) ratio of the transistors is approximately maintained in both inverters, the overall device dimensions in INV2 are much larger. When the width and length of a transistor increase, its parasitic capacitances also increase because these capacitances depend on the physical size of the device. In the case of INV2, these parasitic capacitances become quite significant and add to the capacitance seen at the output node. As a result, the total capacitance that needs to be charged and discharged becomes larger. Since the propagation delay of an inverter mainly depends on how quickly this output capacitance can charge and discharge, the larger effective capacitance in INV2 causes the output to switch more slowly. Therefore, even though INV2 uses larger devices, the increased parasitic capacitances dominate and lead to larger propagation delays compared to INV1.

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A7.

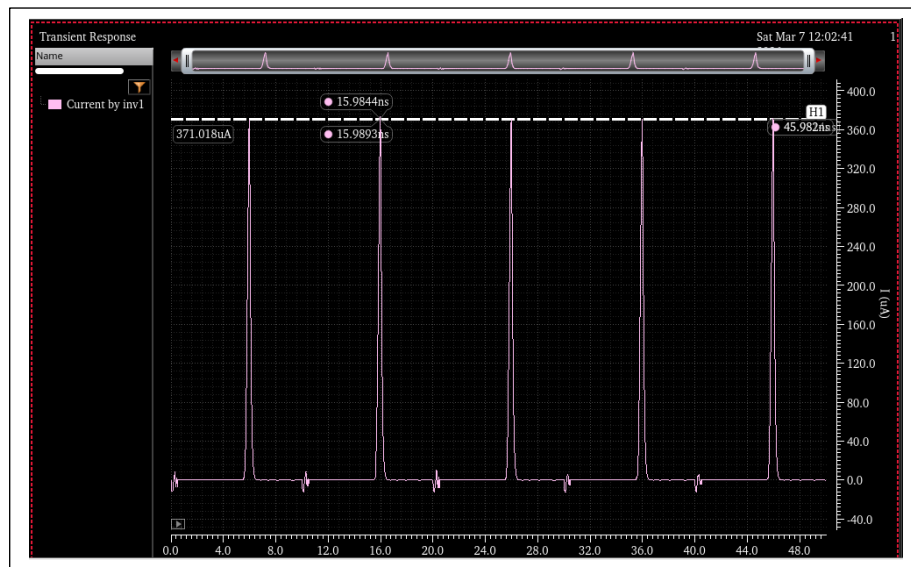


Fig. 6

Explanation-7:

The average power consumed by the inverter was obtained from the simulated supply current waveform when the circuit was driven by a 100 MHz clock signal with a load capacitance of 0.05 pF. By taking the average value of the supply current and multiplying it by the supply voltage of 1.8 V, the total simulated power was found to be approximately 18.78 uW.

For comparison, the dynamic power was also estimated using the theoretical expression ($P = CL VDD^2 f_{clk}$). Using ($CL = 0.05$) pF, ($VDD = 1.8$) V, and ($f_{clk} = 100$) MHz, the calculated theoretical power is about 16.2 uW.

The simulated value is slightly higher than the theoretical estimate because the simple formula only considers the power required to charge and discharge the external load capacitance. In the actual circuit, additional factors contribute to the total power consumption. During the input transitions, both the PMOS and NMOS conduct for a short duration, creating a temporary current path from the supply to ground. This results in short-circuit current. In addition, the inverter must also charge and discharge the internal parasitic capacitances of the transistors. These extra components of power are included in the simulation, which is why the simulated power is slightly higher than the theoretical value.

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Explanation-8:

When the clock frequency is increased to 200 MHz, the simulated power consumption increases to about 37.09 μ W. Earlier, at 100 MHz, the simulated power was around 18.78 μ W. Since dynamic power in CMOS circuits is directly proportional to the switching frequency, doubling the frequency is expected to approximately double the power consumption. If the earlier power value is doubled, it gives about 37.56 μ W, which is very close to the simulated value of 37.09 μ W.

The theoretical dynamic power also follows the same trend. When the frequency increases from 100 MHz to 200 MHz, the theoretical power increases from about 16.2 μ W to 32.4 μ W. The simulated power is slightly higher than the theoretical estimate because the simple formula only considers the external load capacitance. In practice, additional power is consumed due to short-circuit current during switching and the charging and discharging of internal parasitic capacitances.

Explanation-9:

When the input rise and fall time is increased to 1 ns, the simulated power consumption is about 20.13 μ W. The dynamic power remains almost the same because the load capacitance, supply voltage, and clock frequency are unchanged. Using the relation ($P = CL VDD^2 f_{clk}$), the dynamic power is still approximately 16.2 μ W.

However, increasing the rise and fall time makes the input signal change more slowly. Because of this slower transition, both the PMOS and NMOS transistors stay ON together for a longer time during switching. This creates a short path from the supply to ground, which increases the short-circuit current.

Due to this additional short-circuit current, the total power becomes slightly higher than the ideal dynamic power value. Even so, the power is still lower than the case when the frequency was increased to 200 MHz, where the power was about 37.09 μ W.

Marking Scheme:

	Rubric	Marks
a	Fig. (1)	2: Clean schematic
		0: Incorrect
b	Fig. (2)	5: Correct
		2: Correct, but unclear plots
		0: Incorrect or plot not readable
c	Explanation (1)	2: Correct observations
		0: Ramblings without a coherent thought
d	Fig. (3)	2: Clean schematic
		0: Incorrect or plot not readable
e	Fig. (4)	5: Correct
		2: Correct, but unclear plots,
		0: Incorrect or plot not readable
f	Explanation (2)	2: Correct observations
		0: Ramblings without a coherent thought
g	Fig. (5)	3: Correct
		1: Correct, but unclear plots,
		0: Incorrect or plot not readable
h	Explanation (3)	2: Correct observations
		0: Ramblings without a coherent thought
i	Table-1	4: All values tabulated clearly
		0: Incorrect
j	Explanation (4)	5: Correct
		3: Decent observations
		0: Ramblings without a coherent thought
k	Table-2	2: All values tabulated clearly for different rise/fall times
		0: Incorrect
		2: All values tabulated clearly for different Load Capacitances
		0: Incorrect
l	Explanation (5)	5: Correct
		3: Decent observations
		0: Ramblings without a coherent thought
m	Table-3	2: All values tabulated clearly
		0: Incorrect
n	Explanation (6)	2: Correct observations
		0: Ramblings without a coherent thought
o	Fig. (6)	5: Correct
		2: Correct, but unclear plots,
		0: Incorrect or plot not readable
p	Explanation (7)	5: Correct
		3: Decent observations
		0: Ramblings without a coherent thought
q	Explanation (8)	5: Correct
		3: Decent observations
		0: Ramblings without a coherent thought
r	Explanation (9)	5: Correct
		3: Decent observations
		0: Ramblings without a coherent thought
s	Late Submission Deductions	0: On-time submission
		-20: Submission within 2 hrs of deadline
		Submission not evaluated: Otherwise
t	Plagiarism Policy	0: Own work
		-65: Copied work
	Total	sum(a:u)
	Maximum score	65

